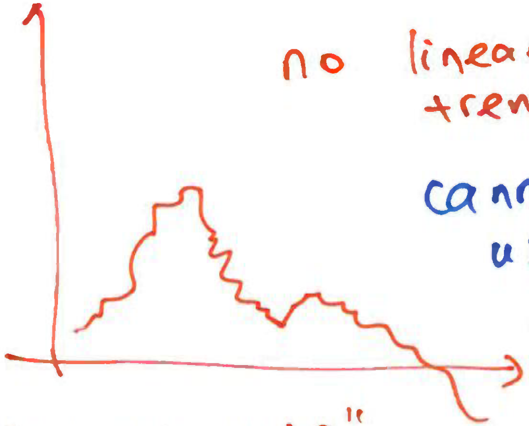
① R.W

$P_t = P_{t-1} + \varepsilon_t$



no linear trend

cannot use AR(1)

"interest rate"

② R.W drift

$P_t = \mu + P_{t-1} + \varepsilon_t$

linear trend.

$\mu > 0$

ln (SP500)

RW
RW drift } N-S. $var(P_t) = \sigma^2 t$

↓

unit root

AR(1) $\quad y_t = \phi y_{t-1} + \varepsilon_t$ ⇒ Y_t must be stationary.

⇒ Y_t estimate AR(1) ⇒ predict Y_t using Y_{t-1}

$|\phi| < 1$

Y_t is R.W (drift) Y_t cannot be estimated AR(1)

$$F_{t-1} = \underset{\substack{\downarrow \\ 5\%}}{Y_{t-1}},\ Y_{t-2},\ Y_{t-3}$$

① $E(Y_{t-1} \mid F_{t-1}) = Y_{t-1} = 5\%$

② $E(Y_{t-1}) = E(Y_t) \Rightarrow$ stationary.

③ $var(\underset{\substack{\| \\ 5\%}}{Y_{t-1}} \mid F_{t-1}) = 0$

④ $var(Y_{t-1}) = var(Y_t) \Rightarrow$ stationary.

$\boxed{vs}$ $E(Y_t \mid F_{t-1}) = c + \phi_1 Y_{t-1} \Rightarrow$ R.V.

$E(Y_t) = \dfrac{c}{1-\phi_1}$ = a number Not a R.V.